

ICE CREAM IN A BAG LAB — M5

Chemistry Worksheet • Partner Lab • 20 points

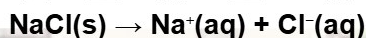
Name(s): _____

Section: _____

Date: _____

Partner(s): _____

LAB CHEMISTRY MODELS / EQUATIONS



The sucrose combustion equation is quantitative practice using the sugar you measured; combustion does NOT happen while making the ice cream.

MEASUREMENTS FOR ONE PAIR

Ingredient / material	Amount
Milk	1/2 cup (120 mL)
Heavy cream	1/2 cup (120 mL)
Sugar (sucrose)	2 tbsp (about 25.0 g)
Vanilla extract	1/2 tsp (2.5 mL)
Ice	about 3 cups (about 700 mL)
Salt (NaCl)	1/3 cup (use 90.0 g for calculations)

PROCEDURE

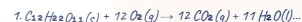
- Measure 120 mL milk + 120 mL heavy cream + 25.0 g sugar + 2.5 mL vanilla into the SMALL bag. Seal tightly.
- Place about 700 mL ice + 90.0 g salt into the LARGE bag.
- Put the sealed small bag inside the large bag. Shake/turn continuously until the mixture thickens.
- Record observations. Keep salt water outside the food bag; wipe the small bag before opening.

OBSERVATIONS

Stage	Texture / appearance	Coldness / temperature	Approx. time
Before shaking	<i>The mixture is liquid with a milky white color. The heavy cream and milk combine to form a smooth, homogeneous liquid. The sugar dissolves into the liquid mixture. The vanilla extract adds a slight brownish tint. Overall, the mixture appears as a thin, pourable white liquid with no visible solid particles remaining.</i>	<i>The mixture feels cold to the touch since the small bag has just been sealed with all ingredients mixed together, and it has not yet been placed inside the large bag with ice and salt. The temperature is cool but not freezing, probably around 4-5°C (refrigerator temperature). The ingredients have not yet been exposed to the ice and salt, so they are at normal/cold beverage temperature. The mixture is noticeably cold but still completely liquid.</i>	<i>0 minutes. This is the starting point before any shaking or freezing has occurred. The small bag has just been sealed with all ingredients mixed together, and it has not yet been placed inside the large bag with ice and salt. No time has elapsed in the ice cream making process.</i>
Halfway			
Finished			

QUANTITATIVE + MOLECULAR ANALYSIS

1. Balance the sucrose combustion equation above using the smallest whole-number coefficients. [2 pt]



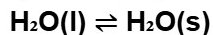
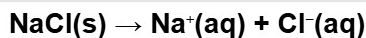
2. Your group used 25.0 g sucrose. Calculate moles of sucrose. Use M = 342.30 g/mol and show units. [3 pt]

3. Using your balanced equation and Question 2, calculate the theoretical moles of CO₂ that would form if that measured sugar were completely combusted. [3 pt]

$$\begin{aligned} 2. n &= \text{mass} \div \text{molar mass} \\ n &= 25.0 \text{ g} \div 342.30 \text{ g/mol} \\ n &= 0.0730 \text{ mol sucrose} \end{aligned}$$

$$\begin{aligned} 3. \text{ From balanced equation: } 1 \text{ mol C}_{12}\text{H}_{22}\text{O}_{11} &\rightarrow 12 \text{ mol CO}_2 \\ n(\text{CO}_2) &= 0.0730 \text{ mol sucrose} \times (12 \text{ mol CO}_2 / 1 \text{ mol sucrose}) \\ n(\text{CO}_2) &= 0.876 \text{ mol CO}_2 \end{aligned}$$

LAB CHEMISTRY MODELS / EQUATIONS



Use the outside ice-salt bath and the actual lab measurements below.

4. Assume the 1/3 cup salt used in the large bag had a mass of 90.0 g. Calculate moles of NaCl using 58.44 g/mol. [2 pt]

5. Each mole of NaCl produces approximately 1 mol Na⁺ and 1 mol Cl⁻. Using Question 4, calculate the total moles of dissolved ions produced. [2 pt]

6. Explain why dissolved salt lowers the freezing point of the ice-water mixture and allows the milk/cream mixture to cool enough to freeze. [2 pt]

7. Milk/cream and dissolved sugar contain polar molecules. Identify one important intermolecular force in the mixture and explain what cooling does to molecular motion and texture. [2 pt]

8. Vanilla extract was only 2.5 mL. Is vanilla acting mainly as a reactant, a solvent/solute component, or a flavor mixture in this lab? Explain why there is no new 'vanilla product' in the equation. [1 pt]

9. If a group doubled the sugar but kept the milk, cream, vanilla, ice, and salt the same, predict one likely effect on sweetness and one likely effect on freezing/texture. Explain. [2 pt]

10. Choose ONE controlled variable (milk/cream amount, sugar, vanilla, ice, salt, or shaking time) and explain why keeping it constant makes the experiment more reliable. [1 pt]

Participation: 20 points | Completed worksheet: 20 points | Total lab score: 40 points

4. moles = mass ÷ molar mass

moles = 90.0 g ÷ 58.44 g/mol = 1.54 mol NaCl

5. Each NaCl gives 2 ions (Na⁺ and Cl⁻)

Total ions = 1.54 mol × 2 = 3.08 mol dissolved ions

6. Dissolved salt lowers the freezing point of water through freezing point depression. The ions disrupt ice crystal formation, so the ice-water bath gets colder than 0°C, allowing the milk mixture to freeze.

8. Vanilla acts mainly as a flavor mixture. The small 2.5 mL amount only adds taste to the product. There is no new vanilla product because it undergoes no chemical reaction, just physical mixing.

Doubling sugar would make ice cream sweeter and lower the freezing point more, causing softer texture or longer freezing time since more solute interferes with solidification.

Keeping salt constant controls the bath temperature. If salt varied, different groups would freeze at different rates, making fair comparison of other variables impossible.